

# Unified Cauer Ladder + Surface Impedance Network via Turning-Point Basis Switch

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**Abstract**—A truncated Cauer Ladder Network (CLN) is seamed to a Surface Impedance Boundary Condition (SIBC) element ( $\alpha = 1/2$  Constant Phase Element) at a turning point  $\omega_N$  given by Matsuo’s corner-frequency formula, with the SIBC scaling fixed by continuity (no fitting). Formally this is the  $s_\infty$  expansion point of the multiple-expansion-points CLN framework. Analytical verification on a Cu slab reproduces the Stoll/Wall closed form across nine decades with ten Cauer rungs and one SIBC element; on a 3D Cu cuboid, an mpmath single-element VIM with HDiv divergence-free hex basis and an NGSolve HCurl Tanimoto-AT extraction yield matching unified  $Y(j\omega)$  from two-to-three bulk Cauer rungs plus the SIBC element, cross-validated against a direct frequency-domain AC sweep.

**Index Terms**—Cauer Ladder Network, Surface Impedance, Turning Point, Expansion Points, Eddy Current.

## I. UNIFIED CLN + SIBC

A truncated Cauer Ladder Network (CLN) [1] represents the eddy-current admittance  $Y(s) = 1/Z(s)$  as a continued fraction with rungs  $(R_{2k}, L_{2k+1})$  encoding the Foster eddy-mode spectrum. The reliable rung count is bounded both by Hankel-Padé ill-conditioning ( $\sim 300\times$  noise amplification per stage [7]) and, more fundamentally, by the basis’s spatial resolution – once skin depth  $\delta(\omega) = \sqrt{2/(\omega\mu\sigma)}$  drops below the basis length scale, further rungs are non-physical. In that regime, however, the Surface Impedance Boundary Condition  $Z_s(j\omega) = \sqrt{j\omega\mu/\sigma}$  – a Constant Phase Element of exponent  $\alpha = 1/2$  – becomes exact.

We seam the two representations at a turning point  $\omega_N$  given by Matsuo’s corner-frequency formula [3] evaluated at the last reliable rung,

$$\omega_N = \sqrt{2^{\alpha/2} - 1} R_{2N-2}/L_{2N-1} \approx 0.4350 R_{2N-2}/L_{2N-1}, \quad (1)$$

yielding  $Y_{unified}(j\omega) = 1/Z_{Cauer}^{(N)}(j\omega)$  for  $\omega < \omega_N$  and  $K_{SIBC} \sqrt{\sigma/(j\omega\mu)}$  above, with  $K_{SIBC} = Y_{Cauer}^{(N)}(j\omega_N)/\sqrt{\sigma/(j\omega_N\mu)}$  fixed by continuity (no fitting). This extension is the  $s_\infty$  expansion point of the multiple-expansion-points CLN framework [2]:  $K_\infty/s_\infty \rightarrow \sigma$  degenerates the energy operator to the conductivity inner product, the natural basis for surface-localised modes.

## II. VERIFICATION

**Slab.** For a Cu slab of half-thickness  $c = 1$  mm, Stoll’s  $Y(s) = \sigma c \tanh(u)/u$  with  $u = c\sqrt{s\mu\sigma}$  [4] and Wall’s odd-integer continued fraction [5] provide the analytical ladder. Feeding the Mittag-Leffler Foster spectrum into our mpmath QD-Wynn extractor at 60-digit precision recovers ten rungs with  $k_L = k_R = 0$  [3] and  $\alpha = 0.500$  to all reported digits;  $\omega_N$  at  $N = 10$  gives  $f_N = 1.12$  MHz,  $K_{SIBC} = 0.937$ , and  $Y_{unified}$  matches  $Y_{full}$  to under 10% at the seam and 1% in the deep Cauer/SIBC regimes across DC to  $10^8$  Hz.

**3D Cu cuboid** (17.72×17.72×2 mm). A pure-mpmath single-element Volume Integral Method (VIM) with HDiv divergence-free hex basis [6] at  $p = 3$  (81 dof) yields two reliable Cauer rungs  $\tau_{pair} = 194.66, 69.96 \mu s$  and a third  $19.40 \mu s$  that lies in the basis noise floor; the *mesh-natural* truncation seams the SIBC element at  $f_N = 9.9 \times 10^2$  Hz. An NGSolve HCurl Tanimoto-AT extraction produces an extra leading stage corresponding to the  $t=0^+$  skin-layer impulse response; identifying

it as the SIBC seed and treating stages  $\{1, 2, 3\}$  as the bulk Cauer ladder yields three rungs  $\tau_{pair} = 218.99, 84.16, 58.03 \mu s$  that are *bit-identical* across  $h = 0.4$ – $0.5$  mm at  $p = 3$  and  $p = 3$ – $4$  at fixed mesh, confirming mesh and order convergence. The third FEM rung  $L_5 = 9.43 \times 10^{-8}$  provides the seam  $f_N = 1.19$  kHz; the first two FEM rungs agree with VIM within 12–20%. The unified  $Y(j\omega)$  from VIM (two rungs,  $f_N = 990$  Hz) and FEM (three rungs,  $f_N = 1190$  Hz) overlap to graphical precision (Fig. 1). A direct NGSolve frequency-domain AC sweep ( $A_{red} \cdot t = 0$  Dirichlet, 20 frequencies 1 Hz– $10^7$  Hz) provides the ground-truth  $Y_{AC}(j\omega) = |A_{ext}|^2 + \langle A_{red}, A_{ext} \rangle$ , which coincides with the unified prediction throughout the Cauer regime (1 kHz) and decays faster than the SIBC seam above it, consistent with the finite-conductor Faraday-shielding floor.

## REFERENCES

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iboid 17.72×17.72×2 mm Cu — Cauer + SIBC unified Y (VIM 2 + F

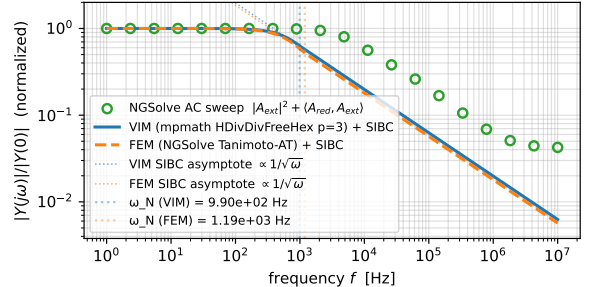


Fig. 1. Cauer + SIBC unified  $|Y(j\omega)|/|Y(0)|$  for the 17.72×17.72×2 mm Cu cuboid in  $B_z$ . VIM (solid, two Cauer rungs at  $p = 3$ , 81 dof) and NGSolve Tanimoto-AT (dashed, three mesh-converged rungs at  $p = 3$ ,  $h = 0.5$  mm) overlap across nine decades. Green circles: independent NGSolve AC sweep ground truth – matches the unified prediction in the Cauer regime and lies slightly below the SIBC seam at high  $\omega$  owing to finite-conductor Faraday shielding.